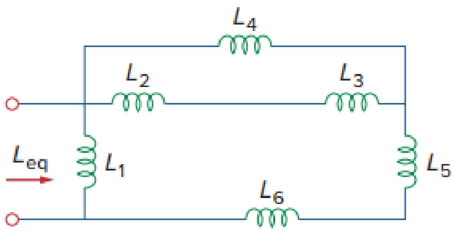
Problem

Using Fig. 6.74, design a problem to help other students better understand how inductors behave when connected in series and when connected in *R* parallel.

Fig. 6.74:



Step-by-step solution

Step 1 of 4

Refer to Figure 6.74 in the textbook.

In the Figure 6.74, the inductors L_2 and L_3 are connected in series. Therefore, their equivalent inductance is,

$$L_{eq1} = L_2 + L_3$$

In the Figure 6.74, the inductors L_6 and L_5 are connected in series. Therefore, their equivalent inductance is,

$$L_{eq2} = L_5 + L_6$$

Step 2 of 4

Draw the modified circuit diagram.

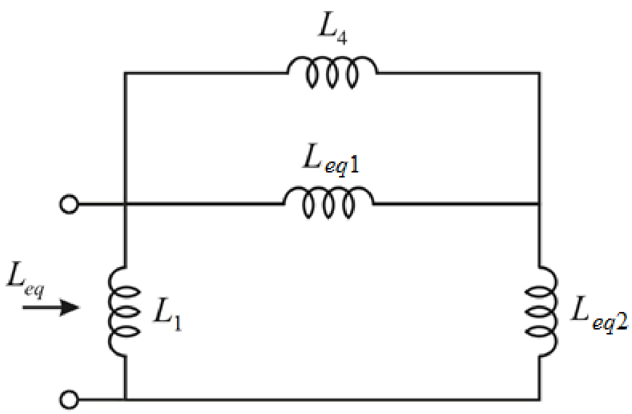


Figure 1

Step 3 of 4

In Figure 1, the inductors L_4 and L_{eq1} are connected in parallel. Therefore, their equivalent inductance is,

$$\begin{aligned} L_{eq3} &= L_4 \parallel L_{eq1} \\ &= \frac{(L_4)(L_{eq1})}{L_4 + L_{eq1}} \\ &= \frac{(L_4)(L_2 + L_3)}{L_4 + L_2 + L_3} \end{aligned}$$

Draw the modified ciruict diagram of Figure 1.

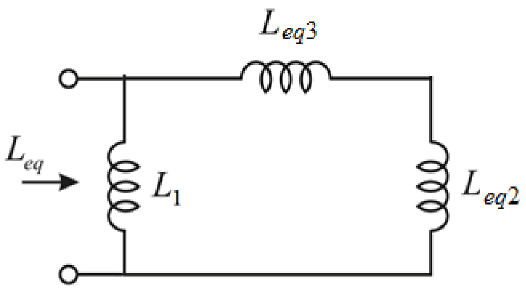


Figure 2

Step 4 of 4

From Figure 2, the equivalent inductance is,

$$\begin{aligned} L_{eq} &= (L_{eq2} + L_{eq3}) \parallel L_1 \\ &= \frac{(L_{eq2} + L_{eq3})(L_1)}{L_{eq2} + L_{eq3} + L_1} \\ &= \frac{\left(L_5 + L_6 + \frac{(L_4)(L_2 + L_3)}{L_4 + L_2 + L_3}\right)(L_1)}{L_5 + L_6 + \frac{(L_4)(L_2 + L_3)}{L_4 + L_2 + L_3} + L_1} \\ &= \frac{((L_5 + L_6)(L_4 + L_2 + L_3) + (L_4)(L_2 + L_3))(L_1)}{(L_5 + L_6 + L_1)(L_4 + L_2 + L_3) + (L_4)(L_2 + L_3)} \end{aligned}$$

$$= \frac{L_1[(L_5 + L_6)(L_2 + L_3 + L_4) + (L_4)(L_2 + L_3)]}{(L_1 + L_5 + L_6)(L_2 + L_3 + L_4) + (L_4)(L_2 + L_3)}$$

Therefore, the equivalent inductance, L_{eq} is

$$\boxed{\frac{L_1[(L_5 + L_6)(L_2 + L_3 + L_4) + (L_4)(L_2 + L_3)]}{(L_1 + L_5 + L_6)(L_2 + L_3 + L_4) + (L_4)(L_2 + L_3)}}.$$